

CENG-322 Deliverable 2

Team name:- Vision Assist Innovators

Project Name:- Smart Assistive System for Visually Impaired People

Group Number:- 05

Student's Names and IDs

- Sarang Prajapati – N01662036
- Krish Patel – N01666556
- Kush Patel – N01657387
- Daksh Rana – N01664095

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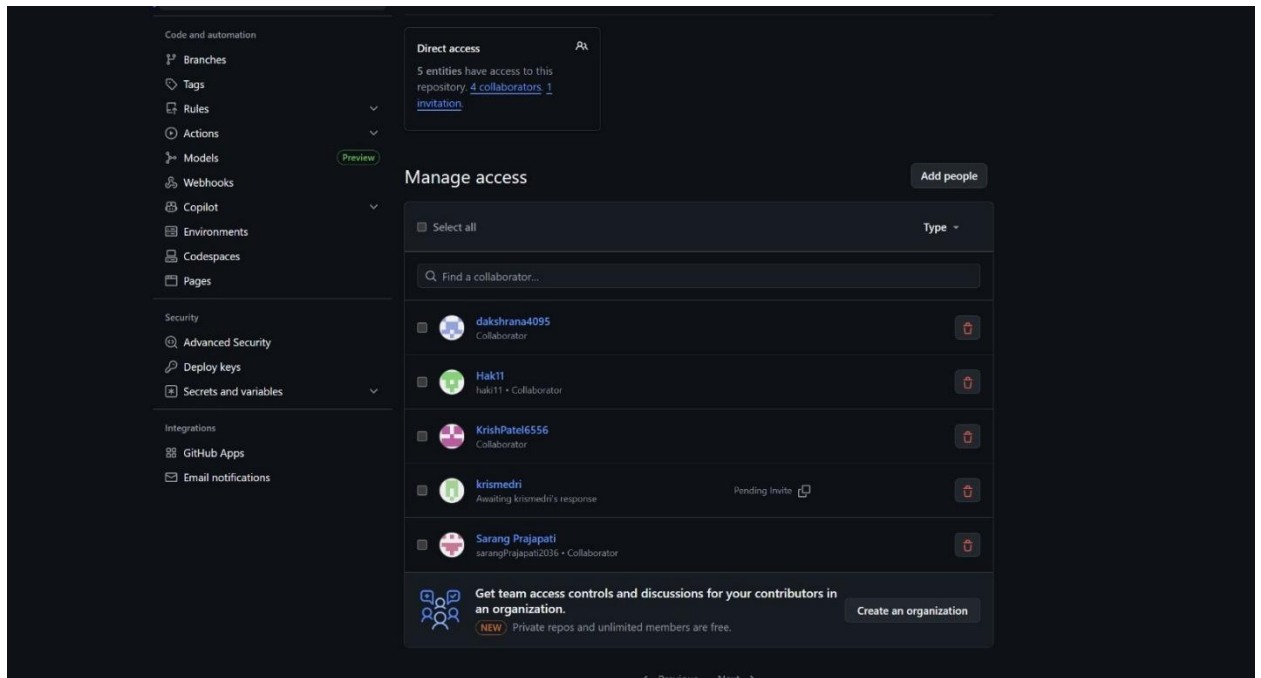
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Name	Student ID	GitHub ID	Signature	Effort
Sarang Prajapati	N01662036	SarangPrajapati2036	Sarang	100%
Krish Patel	N01666556	KrishPatel6556	Krish	100%
Kush Patel	N01657387	PatelKush7387	Kush	100%
Daksh Rana	N01664095	DakshRana4095	Rana	100%

We all contributed equally and ensured an efficient distribution of workload. Additionally, we supported each other by completing tasks that some team members could not finish due to obstacles in their schedules.

Requirements (Items 1-25)

- 1) In the PDF, write the requirement and comment on it for each item. Include the question and the answer.
Answer: We wrote each requirement here with proper explanation.
- 2) Project Scope and goals that are targeted in this course.
Answer: Scope: App with sensors to help blind people walk safe. Uses light, color, distance sensors. Sends alerts by sound or shake. Goals: Build Android app, use agile,(scrum) add hardware(at last), learn UI and DB.
- 3) GitHub Repo link and strategy. All members must contribute to the repo. Explain the GitHub strategy.
Answer: Link: <https://github.com/KushPatel7387/SmartAssistiveSystem>.
Strategy: We used the main branch for finalized code. Each team member first pulled the latest version of the project before committing their own changes. Communication was maintained among all members to ensure everyone pulled the latest updates before starting their work. Although the beginning was a bit challenging—with multiple errors and several branches being created—we eventually resolved the issues and streamlined our workflow successfully.
- 4) Verification of Link is working :- Yes , the link is working.
- 5) Screenshot of GitHub Project Invitation to our hardware Professor(Kristian Medri)



- 6) Each Member should have 5 minimum commits:- Yes, all team members were successfully able to have 5 commits on the master branch.
- 7) . Marks deducted for members who are not meeting the minimum 5 commits (i.e. if only 4 commits, 20% percent deducted, ...etc.).
- 8) Minimum of 5 stories and each story must be split to minimum of 4 tasks. Use one of the online tools to build the stories and tasks, i.e. use Trello (<https://trello.com/en>) or Monday (<https://monday.com/>) or any other appropriate tool i.e. <https://www.notion.com/> <https://clickup.com/>
 Answer:- Initially, we created the Scrum dashboard on Trello. However, based on your requirements, we realized that the Trello board followed a Kanban format. Therefore, we later switched to Monday.com to properly implement the Scrum framework.
- 9) Take a screenshot showing clearly the stories and breakdown of tasks with owner, status, start/end date, size, and priority.
 Answer: We made the Dashboard on Monday.com with 5 stories and with some subtasks along with due date , priority and status.

Smart Assistive System ▾

Main table ... +

New task ▾

🔍 Search

👤 Person

🔼 Filter ▾

⬆️ Sort

🔍 Hide

📅 Group by ...

▾ Story 1: Database Setup

☐	Task		Owner	Status ⓘ	Due date ⓘ	Priority	+
☐	Design DB schema on paper	✦ Ⓢ	👤	Done	🕒 Sep-30	Low	
☐	Create DB instance on cloud (Fireb...	✦ 🗨️	👤 KP	Done	✓ Oct-5	Low	
☐	Connect Android app to DB	✦ Ⓢ	👤 KP	Working on it	🕒 Oct 6	Low	
☐	Test data insertion/retrieval	✦ Ⓢ	👤 KP +2	Not Started	🕒 Oct 5	High	
☐	Test CRUD operations with sample ...	✦ Ⓢ	👤	Working on it		Medium	
☐	+ Add task						

Sep 30 - Oct...

▾ Story 2: Sensor Integration

☐	Task		Owner	Status ⓘ	Due date ⓘ	Priority	+
☐	Interface AS7262 Visible Sensor	✦ Ⓢ	👤	Not Started	🕒 Dec 31	Low	
☐	Interface TSL2591 Light Sensor	✦ 🗨️	👤 KP	Not Started	🕒 Dec 31	Low	
☐	Test sensor data with Android app ...	✦ Ⓢ	👤 KP KP	Not Started	🕒 Dec 31	Low	
☐	Calibrate readings, store results in ...	✦ Ⓢ	👤 KP	Not Started	🕒 Dec 19	Low	
☐	Document calibration constants & ...	✦ Ⓢ	👤 KP	Not Started	🕒 Dec 19	Low	
☐	Placing order for sensors	✦ Ⓢ	👤 KP +3	Done	🕒 Oct-4	High	
☐	Integrate TCS34725 Color Sensor	✦ 🗨️	👤	Not Started	🕒 Dec 15	Low	
☐	Integrate VL53L1X Distance Sensor	✦ 🗨️	👤 KP	Not Started	🕒 Nov 15	Low	
☐	+ Add task						

Oct 1 - Dec 31

12. Tasks which are marked done, explain how did you meet the DoD criteria.

Answer: For the Splash task, the code was completed, checked, and tested on the phone. Everything worked fine with no errors. The task was reviewed and merged into the main branch, so it met all the DoD points.

List of our DoD:-

- Ordering the sensors respectively.
- Code review,
- Database Setup
- Business Model Canvas
- Scrum Dashboard
- Gant Chart

- Merge to main.
- No open problems.

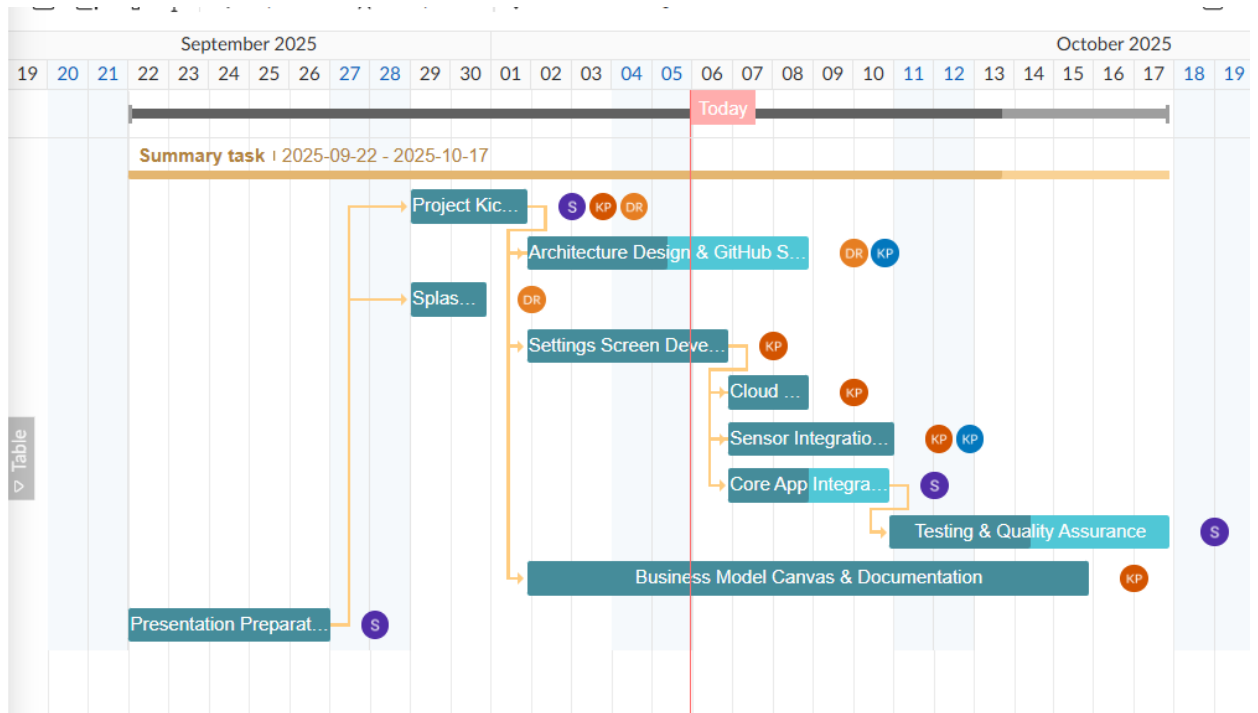
13. Business Model Canvas, all the 9 fields are clearly shown and explained.

Customer Segments • Visually impaired individuals • Elderly people with mobility challenges • Caregivers & families • NGOs and healthcare centers	Value Proposition • Affordable assistive navigation device • Combines sensors for light, color, distance, obstacles. • Real-time feedback through mobile app. • Customizable alerts for user needs.	Channels • Mobile application (Android/iOS) • Online distribution (website/app stores). • Partnerships with healthcare or rehabilitation centers. • NGOs for disability support.
Customer Relationships • Easy-to-use interface & training • Continuous support (FAQ, in-app help) • Personalized alerts/settings • Feedback System for improvements	Key Resources • Hardware: Raspberry Pi, Sensors (TSL2591, TCS34725, VL53L1X, Ultrasonic) • Cloud Database (MongoDB Atlas / Firebase) • Mobile App (for alerts & control) • Development team (you & group members)	Revenue Streams • Device sales (one-time hardware purchase) • Premium mobile app features (subscription) • Partnerships with organizations (funding, grants) • Future: maintenance/service plans
Key Activities • Sensor integration & testing • Mobile app development • Database setup on the cloud • Deployment & user trials • Regular updates & maintenance	Key Partners • Sensor suppliers / hardware vendors • Cloud service providers (MongoDB, Firebase, AWS) • Accessibility/Disability support organizations • Academic institutions for testing	Cost Structure • Hardware components (sensors, Raspberry Pi) • Cloud hosting & database costs • App development & testing effort • Training & documentation • Marketing/distribution (if productized)

14. Gantt Chart:-

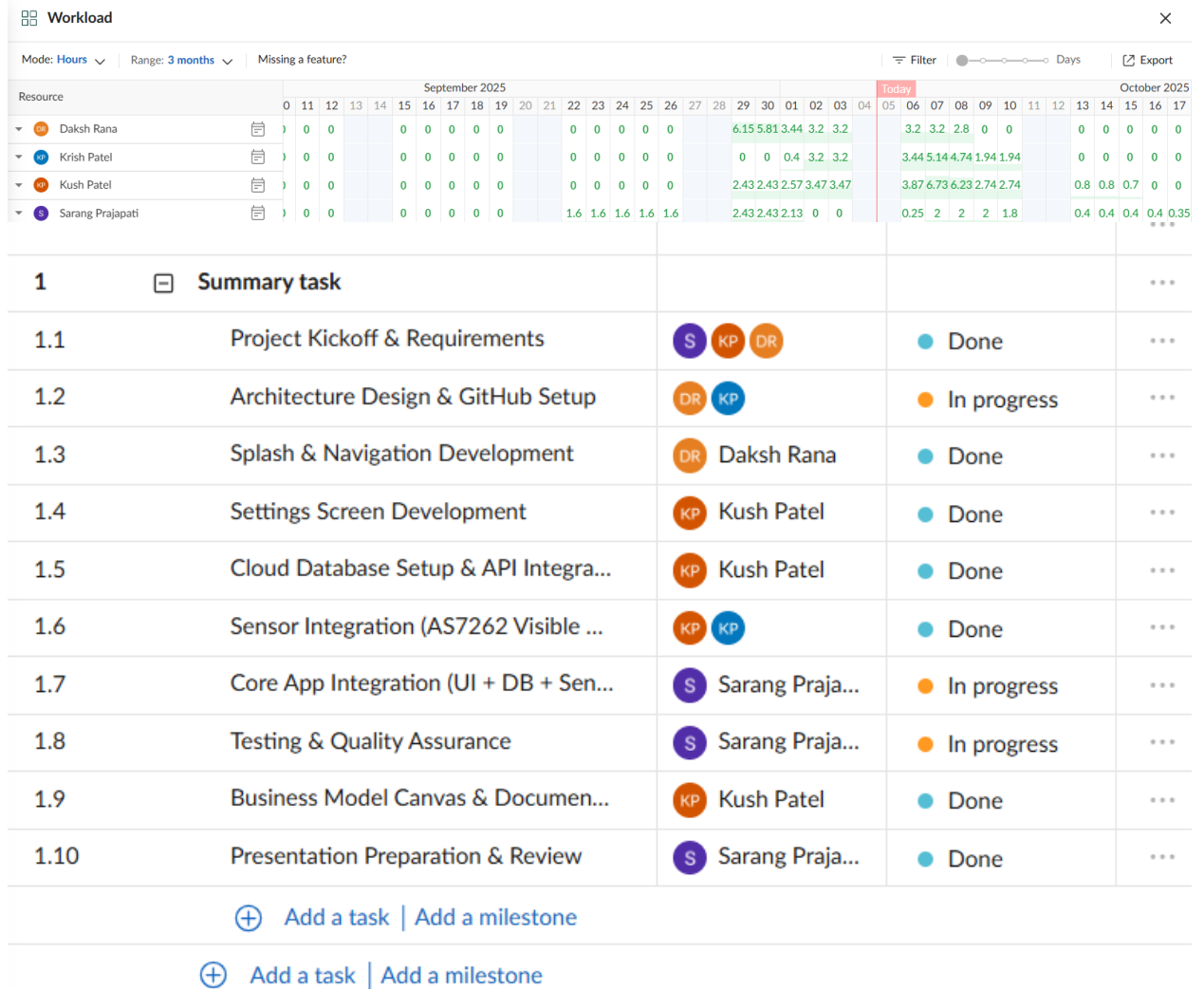
15. The Gantt chart must show the work for a minimum of 10 components and the timelines..Answer: 11 parts: Planning, Repo, UI, Splash, Settings, Language,, Alerts, Safety, Sensors, Docs. Timelines from Sept 22 to Dec 15.

16. Gantt chart should show the project from start to end. Answer: Starts Sept 26, 2025.

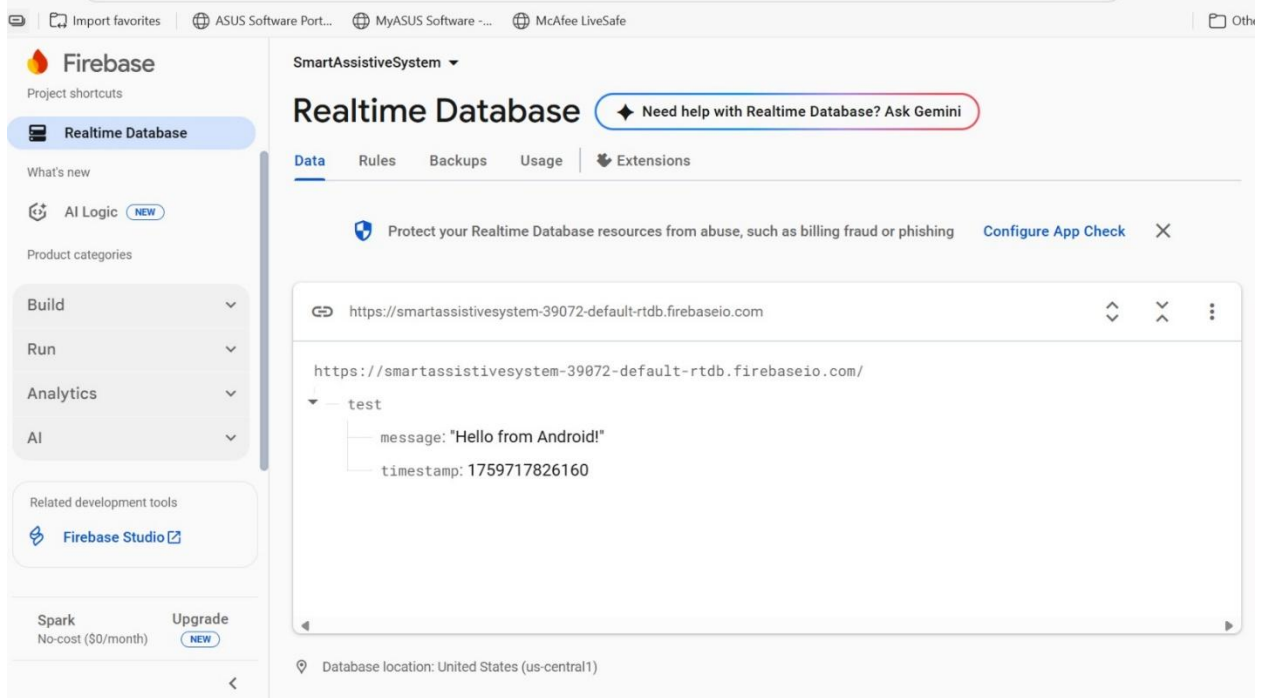


17. Gantt chart is different than your Scrum Dashboard... Answer: Gantt for big steps.
Dashboard for small tasks.

18. Multiple screenshots of the Gantt charts showing the details.



19. Screenshot showing you have created the DB on the cloud.



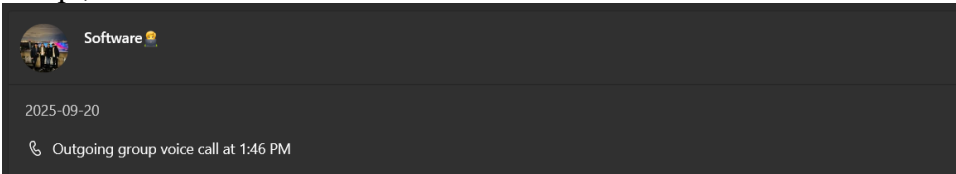
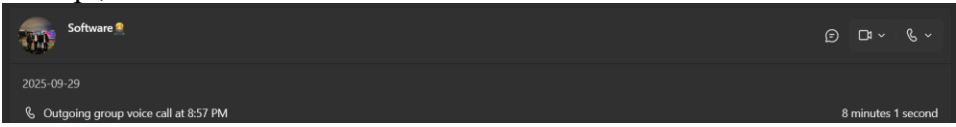
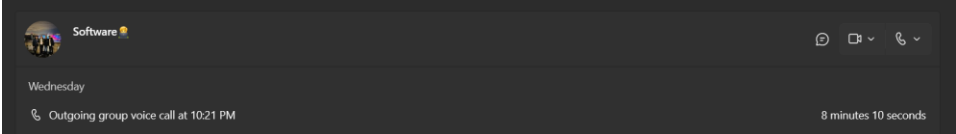
20. Explain how you are planning to use the DB.

Answer: We are planning to use the Firebase Realtime Database to store and update data from our application in real time. It will keep all user or device information synced instantly, so any changes made will be seen by all users right away. The database will also be used to save readings, user inputs, and other important data that our app needs to show or process live.

21. Coding work progress since deliverable 1. What additional features/functionality added since deliverable 1

Answer: Since Deliverable 1, we added runtime functionalities, made several UI improvements, and added images to make the app more user-friendly. We also removed all hardcoded texts, committed our code on time, and applied a consistent theme across the app.

22. A Table records the daily stand-ups outcome.

Date	Info
9 Sept, 2025 	Just a small talk about the overview of tasks which were needed to done.
29 Sept, 2025 	Dashboard, Gantt Chart , Business Model Canva(completed) plus Subtask were distributed .
Oct 5, 2025 	Deliverable 2 :- things which were left to do.